

1 Intersections

1.1 Resultants

Lemma 1.1.1. For R a UFD, and $f, g \in R[x]$ of degree $d \geq 1$ and $e \geq 1$ respectively. f and g have a non-constant common factor iff there exists non-zero polynomials $a, b \in R[x]$ such that:

$$af + bg = 0$$

where $\deg(a) \leq e - 1$ and $\deg(b) \leq d - 1$.

Proof. Suppose f, g have a common factor h of degree $n \geq 1$, then there exists $p, q \in R[x]$ such that $f = ph$ and $g = qh$, so we can have:

$$qf - pg = 0$$

where $\deg(q) = e - n \leq e - 1$ and $\deg(p) = d - n \leq d - 1$.

If, instead, we can find such $a, b \in R[x]$ then by factoring af into irreducible components:

$$af = c \cdot h_1^{\alpha_1} h_2^{\alpha_2} \cdots h_k^{\alpha_k} = -bg$$

if f and g contain none of the same factors, g must divide a , which contradicts the degree requirement. $\square$

This tells us that the polynomials f, g are coprime iff the polynomials:

$$x^{e-1}f, \dots, xf, f, x^{d-1}g, \dots, xg, g$$

are linearly independent. The best test we know for linear independence is the determinant, leading us to the definition of the resultant.

Definition 1.1.2. For R a UFD, and $f, g \in R[x]$ of the form:

$$f(x) = a_d x^d + \cdots + a_1 x + a_0 \quad g(x) = b_e x^e + \cdots + b_1 x + b_0$$

where $a_d, b_e \neq 0$, the **resultant** of f, g is:

$$\mathcal{R}_{fg} := \begin{vmatrix} a_d & a_{d-1} & a_{d-2} & \cdots & a_0 & & & & & \\ & a_d & a_{d-1} & \cdots & a_1 & a_0 & & & & \\ & & a_d & \cdots & a_2 & a_1 & a_0 & & & \\ & & & \ddots & & \ddots & \ddots & \ddots & & \\ & & & & a_d & \cdots & a_2 & a_1 & a_0 & \\ b_e & b_{e-1} & b_{e-2} & \cdots & b_0 & & & & & \\ & b_e & b_{e-1} & \cdots & b_1 & b_0 & & & & \\ & & b_e & \cdots & b_2 & b_1 & b_0 & & & \\ & & & \ddots & & \ddots & \ddots & \ddots & & \\ & & & & b_e & \cdots & b_2 & b_1 & b_0 & \end{vmatrix} \begin{matrix} \left. \vphantom{\begin{matrix} a_d \\ & a_d \\ & & a_d \\ & & & \ddots \\ & & & & a_d \end{matrix}} \right\} e \text{ rows} \\ \left. \vphantom{\begin{matrix} b_e \\ & b_e \\ & & b_e \\ & & & \ddots \\ & & & & b_e \end{matrix}} \right\} d \text{ rows} \end{matrix}$$

the determinant of this $(d + e) \times (d + e)$ matrix.¹

Corollary 1.1.3. f and g share a non-constant common factor iff $\mathcal{R}_{f,g} = 0$.

We should now consider the resultant in the context of projective plane curves, not just arbitrary polynomials.

¹This is called the **Sylvester** matrix.

1.2 Weak Bézout

Throughout this section, we want to consider two projective plane curves:

$$C = \mathbb{V}(F(x, y, z)) \quad C' = \mathbb{V}(G(x, y, z))$$

For homogeneous polynomials $F, G \in k[x, y, z]$, which we view as $R[x]$ over the UFD $R = k[y, z]$, given by:

$$\begin{aligned} F(x, y, z) &= f_0(y, z)x^d + f_1(y, z)x^{d-1} + \cdots + f_d(y, z) \\ G(x, y, z) &= g_0(y, z)x^e + g_1(y, z)x^{e-1} + \cdots + g_e(y, z) \end{aligned}$$

we know these curves won't cover all of $\mathbb{P}^2$, so they must avoid some common point. By applying a projective transformation, we may assume that they both miss $[1, 0, 0]$.

Proposition 1.2.1. There exist polynomials $A, B \in k[x, y, z]$ with x -degrees less than e, d respectively such that:

$$\mathcal{R}_{F,G}(y, z) = AF + BG \in k[y, z]$$

Furthermore, if $\mathcal{R}_{F,G}$ is nonzero, it will be a homogeneous polynomial of degree de .

Proof. We will perform a series of column operators on the Sylvester matrix: for each $1 \leq k \leq d + e - 1$, add x^{d+e-k} times the k th column to the last column. This doesn't effect the determinant, and gives the final column the form:

$$(x^{e-1}F, \dots, xF, F, x^{d-1}G, \dots, xG, G)$$

we can now expand the determinant along this column, giving the polynomials A, B ; as every other entry in the matrix is in $k[y, z]$ the degrees of A and B are as desired.

Now to show $\mathcal{R}_{F,G}$ is homogeneous, we can write:

$$\mathcal{R}_{F,G} = \det(-) = \sum_{\sigma \in S^{d+e}} (-1)^\sigma \prod_{i=1}^d f_{\sigma(i)-i} \prod_{j=1}^e g_{\sigma(j+e)-j}$$

each of these is a product of homogeneous polynomials of degree:

$$\begin{aligned} \sum_{i=1}^d \deg(f_{\sigma(i)-i}) + \sum_{j=1}^e \deg(g_{\sigma(j+e)-j}) &= \sum_{i=1}^d \sigma(i) - i + \sum_{j=1}^e \sigma(j+e) - j \\ &= \frac{(d+e)(d+e+1)}{2} - \frac{d(d+1)}{2} - \frac{e(e+1)}{2} = de \end{aligned} \quad \square$$

Proposition 1.2.2. $\mathcal{R}_{F,G}(y, z)$ is zero iff F and G share a non-constant common factor in

Proof. We already know $\mathcal{R}_{F,G}$ is zero iff F and G share a non-constant common factor in $k[y, z]$. But, by our assumption that F and G are nonzero on $[1, 0, 0]$ this factor must have positive x -degree, else $F(1, 0, 0)$ has 0 as a factor. $\square$

Theorem 1.2.3 (Weak Bézout theorem). The intersection $C \cap C'$ is nonempty, and either $|C \cap C'| \leq de$ or $C \cap C'$ contains a plane curve.

Proof. $\square$

- 1.3 Conics
- 1.4 Constructive nullstellensatz
- 1.5 Intersection multiplicity
- 1.6 Strong Bézout
- 1.7 Multiple points
- 1.8 Cubics